

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8CJ408 / MAT8MN408			
Course Title	DIFFERENTIAL EQUATIONS			
Type of Course	Major			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic knowledge of calculus of one variable and an introductory course in Real Analysis			
Course Summary	The course enhances the skill to solve ordinary differential equation using specific methods analytically and computationally for first and higher order differential equations. Most of the fundamental phenomena occurring in the nature are expressed as a differential equation. Students must know how to model any physical phenomena using differential equations.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand and apply the existence and uniqueness theorems for second-order differential equations, including methods such as the method of successive approximations and Picard's theorem.	Ap	P	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO2	Analyse and solve second-order differential equations using power series methods, including ordinary points, regular singular points, and specific functions such as Gauss's Hypergeometric Equation and Legendre Polynomials.	An	C	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO3	Evaluate and determine the stability of autonomous systems and critical points for linear and nonlinear systems using the phase plane analysis and Lyapunov's direct method.	E	M	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book		Differential Equations With Applications And Historical Notes, Third Edition,George F. Simmons.		
Module	Unit	Content	Hrs (48+ 12)	Marks Ext: 70
I	Second Order Differential Equations		12	Min.15
		Existence and Uniqueness of Solutions and Power Series method of solving differential equations		
	1	69 Method of Successive Approximations		
	2	70 Picard’s theorem, theorems A& B (proofs are optional).		
	3	71 Systems. The Second Order Equations		
	4	26 Introduction. A review of Power Series		
	5	27 Series solutions of first order equations		
	6	28 Second order Equations. Ordinary points		
	7	29 Regular singular points		
II	Power Series Solutions and Special Functions		11	Min.15
	8	30 Regular Singular Points continued		
	9	31 Gauss’s Hypergeometric Equation		
	10	31 Gauss’s Hypergeometric Equation Reduction to Hypergeometric equation		
	11	32 The Point at Infinity		
	12	44 Legendre Polynomials (proofs of Rodrigues’ formula is optional)		
III	Special Functions (Contd.)		12	Min.15
	13	45 Properties of Legendre Polynomials		
	14	46 Bessel functions.		
	15	46 Bessel functions. The Gamma function		
	16	47 Properties of Bessel functions		
	17	47 Properties of Bessel functions Zeros and Bessel series. Bessel expansions		
IV	Autonomous Systems. Stability of Linear and Nonlinear Systems		13	Min.15
	18	58 Autonomous systems. The phase plane and its phenomena		
	19	59 Types of critical points		
	20	59 Types of critical points. Stability		
	21	60 Critical points and stability for linear system		
	22	61 Stability by lyapunov direct method		
V	Open Ended		12	
	● Proof of Picard’s theorem ● Proof of theorem B of Unit I ● Proof of Rodrigues’ formula for Legendre polynomials ● Analyse solutions of Differential Equations using softwares like Python			

References			
1. G. Birkhoff and G.C. Rota: Ordinary Differential Equations (3rd Edn.); Edn. Wiley & Sons; 1978			
2. W.E. Boyce and R.C. DiPrima: Elementary Differential Equations and boundary value problems (2nd Edn.); John Wiley & Sons, NY; 1969			
3. A. Chakrabarti: Elements of ordinary Differential Equations and special functions; Wiley Eastern Ltd., New Delhi; 1990			
4. E.A. Coddington: An Introduction to Ordinary Differential Equations; Prentice Hall of India, New Delhi; 1974			
5. A. K. Nandakumaran, P. S. Datti, Raju K. George: Ordinary Differential Equations: Principles and Applications, Cambridge University Press			

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	2	2	1	3	0	3	1	3	0	2
CO 2	2	2	1	0	3	0	3	2	3	0	3
CO 3	1	2	2	2	3	0	3	1	3	0	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

ELECTIVE COURSES